



BUYERS' GUIDE

ESTYLE

Selecting An Appropriate ROOFTOP SOLAR SYSTEM



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India is blessed with abundant sunshine for most of the year. Switching to solar energy is therefore an extremely logical option. Solar energy is more economical and environment-friendly as compared to non-renewable sources of energy like coal and petroleum. The government of India has launched training programmes, framed solar net-metering guidelines as well as announced other incentives to promote the use of solar energy.

Selecting the right product is critical for the smooth running of daily operations. This guide will assist you in making an informed decision.

What solar panels are

Solar panels (also called photovoltaics) are primarily rectangular silicon devices that can convert light into electricity. These comprise several solar cells, which are spread over a large area that can work together to offer sufficient power for it to be deemed useful. Efficacy of solar panels depends on how well these face the sun.

Solar cells are available in all sizes, colours and power ratings. Applications that require minimal amount of power (20W to 40W) include solar lamps, lanterns, torches and so on. Rooftop installations require pan-

els with heavier output of 150W to 300W or even more.

These are of the following two types:

Crystalline-silicon panels. These include monocrystalline and multi-crystalline panels. The former produce higher amount of power and are costlier than the latter.

Thin-film panels. These are used in building integrated photovoltaic applications, where these become a part of the building structure.

Factors that determine price

Price of a solar panel can vary from ₹ 30/kW to ₹ 60/kW. An imported module can cost about ₹ 40/kW to ₹ 45/kW. Good ones manufactured in India can cost ₹ 30/kW to ₹ 32/kW (for bulk transactions). Retail price of a module producing 3W - 40W may vary between ₹ 60/kW and ₹ 90/kW and modules producing 50W - 300W between ₹ 55/kW and ₹ 60/kW.

Price of solar panels varies according to type, wattage, brand and testing standards, as explained below.

Type. Thin-film panels cost less than crystalline silicon panels.

Wattage. Panels that have a higher power output have a higher price as these produce more electricity.

SOME SOLAR PANEL BRANDS AND THEIR KEY FEATURES

	SuKam	Tata BP Solar	Vikram	Moser Baer	Alplex
Key features	- Lowest yearly degradation rate - Optical low iron tempered glass for strong permeability and shock resistance - Heavy-duty anodised aluminium frame for enduring high load forces, scratch resistance, corrosion-resistance and weatherproof surface	- High fill factor for improved energy-conversion efficiency - Electroluminescence tested for micro-cracks - Torsion and corrosion-resistant with anodised aluminium frame	- Bloomberg Tier 1* - Positive power output tolerance - Excellent low-light response - Global recognition	- Multi-layered back sheet provides weatherable barrier to modules, high performance in rugged environments around the world, high dielectric performance, superior partial discharge and electrical insulation - Sturdy anodised aluminium frame design with twin wall profile, and water drainage holes ensure superior torsional resistance and longevity in adverse conditions - Bill of materials successfully tested for environmental tests as per IEC-61215 - Exceptional low power tolerance ensures customers actually get the module power outputs they expect	- Robust modules as 4mm glass is used instead of the usual 3.2mm - Green modules manufactured using wind power - Lower structural cost for each MW of capacity owing to low weight of modules

*Bloomberg defines a Tier 1 solar manufacturer as one who has provided products to five different projects, which have been financed non-recourse by five different (non-development) banks in the past two years